

Question 3

Take a number, count how many digits it has and state whether it is a single-digit, double-digit or triple (or more) number.

Pseudocode

Begin

Declare N and count as integer

Take N as input

Set count $\leftarrow$ 0

If $N = 0$

Count $\leftarrow$ 1

else

while $N > 0$

$N \leftarrow N / 10$

count $\leftarrow$ count + 1

end while

end if

If count = 1

Display "Single-digit number"

else if count = 2

Display "Double-digit number"

else if count = 3

Display "Triple-digit number"

else

Display "Four or more digit number"

end if

end

Algorithm

1. start
2. Input a number N .
3. Set $count = 0$
4. If $N = 0$ set $count = 1$.
5. Otherwise, repeat while $N > 0$
 - Divide N by 10 using integer division.
 - Increase $count$ by 1.
6. If $count = 1$, display "Single digit number".
7. Else if $count = 2$, display "Double digit number".
8. Else if $count = 3$, display "Triple-digit number".
9. Else display "4 or more digit number".
10. stop